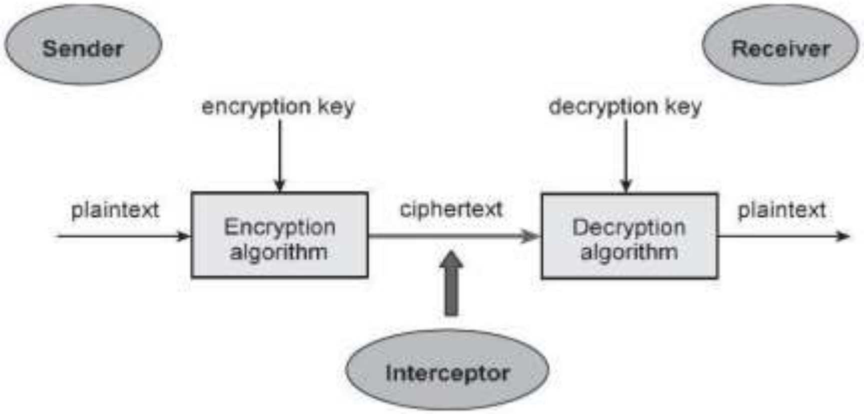
5CM512
Communication and Security Protocols

Week 02: Symmetric Encryption

Cryptographic systems



- Symmetric Cryptography
  - encryption and decryption keys are essentially the same
  - Historical cryptosystems are all symmetric
- Asymmetric (public key) Cryptography
  - Encryption and decryption keys are fundamentally different

Substitution Ciphers

1. The clear text message would be encoded using a key of 3.

Replace alphabet with another alphabet based on key
Example: Caesar’s Cipher

2. Shift the top scroll over by three characters (key of 3), an A becomes D, B becomes E, and so on.

3.The clear text message would be encrypted as follows using a key of 3.

Simple Substitution Cipher

- Caesar’s cipher can be easily broken. 26 possible permutations.
- Instead of shifting the alphabet, the Simple Substitution Cipher allows any permutation of letters of the alphabet
    - For example plaintext ‘point’ will create ciphertext ‘MJBXZ’ using the following permutation

Plaintext Alphabet	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
Ciphertext Alphabet	K	D	G	F	N	S	L	V	B	W	A	H	E	X	J	M	Q	C	P	Z	R	T	Y	I	U	O

    - This allows 26! possible permutations for a 26-letter alphabet. Not bad.
    - Can this still be defeated?

The Vigenère cipher

- Vigenère cipher is an improved substitution cipher
  - uses Caesar's cipher, but changes the shift number based on the location of the alphabet letter in plaintext and a key

- For example:
  - Plaintext: "AARDVARK"
  - Key: "DIG" (Caesar's shift number: "4 9 7")

Plain Text	**A**	**A**	**R**	**D**	**V**	**A**	**R**	**K**
Key	D	I	G	D	I	G	D	I
Cipher text	D	I	X	G	D	G	U	S

  - Makes frequency analysis less effective. The same alphabet character could change to a different cipher pending on its location.
  - Possible key permutations: alphabet length ^ key length
- Could still be defeated if you know the key length

The Vigenère cipher – substitution table

Msg: samplemessage

Key: passwordpassw

CTXT: HAEHHSDHHSSYA

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

One-time Pad

Plain Text	**1**	**1**	**0**	**1**	**0**	**0**	**1**	**1**
Key (Apply)	**0**	**1**	**0**	**1**	**0**	**1**	**0**	**1**
XOR (Cipher Text)	**1**	**0**	**0**	**0**	**0**	**1**	**1**	**0**
Key (Re-Apply)	**0**	**1**	**0**	**1**	**0**	**1**	**0**	**1**
XOR (Plain Text)	**1**	**1**	**0**	**1**	**0**	**0**	**1**	**1**

- Variation of substitution cipher offering perfect secrecy
  - Key generated randomly
  - Key at least as long as the original message
  - Key is used only once
- **Computationally impossible to break by brute-force**

One-Time Pad: Encryption (OTP)

e=000 h=001 i=010 k=011 l=100 r=101 s=110 t=111

Encryption: Plaintext ⊕ Key = Ciphertext

	h	e	i	l	h	i	t	l	e	r
Plaintext:	001	000	010	100	001	010	111	100	000	101
Key:	111	101	110	101	111	100	000	101	110	000
Ciphertext:	110	101	100	001	110	110	111	001	110	101
	s	r	l	h	s	s	t	h	s	r

(Source: Information Security: Principles and Practice)

One-Time Pad: Decryption

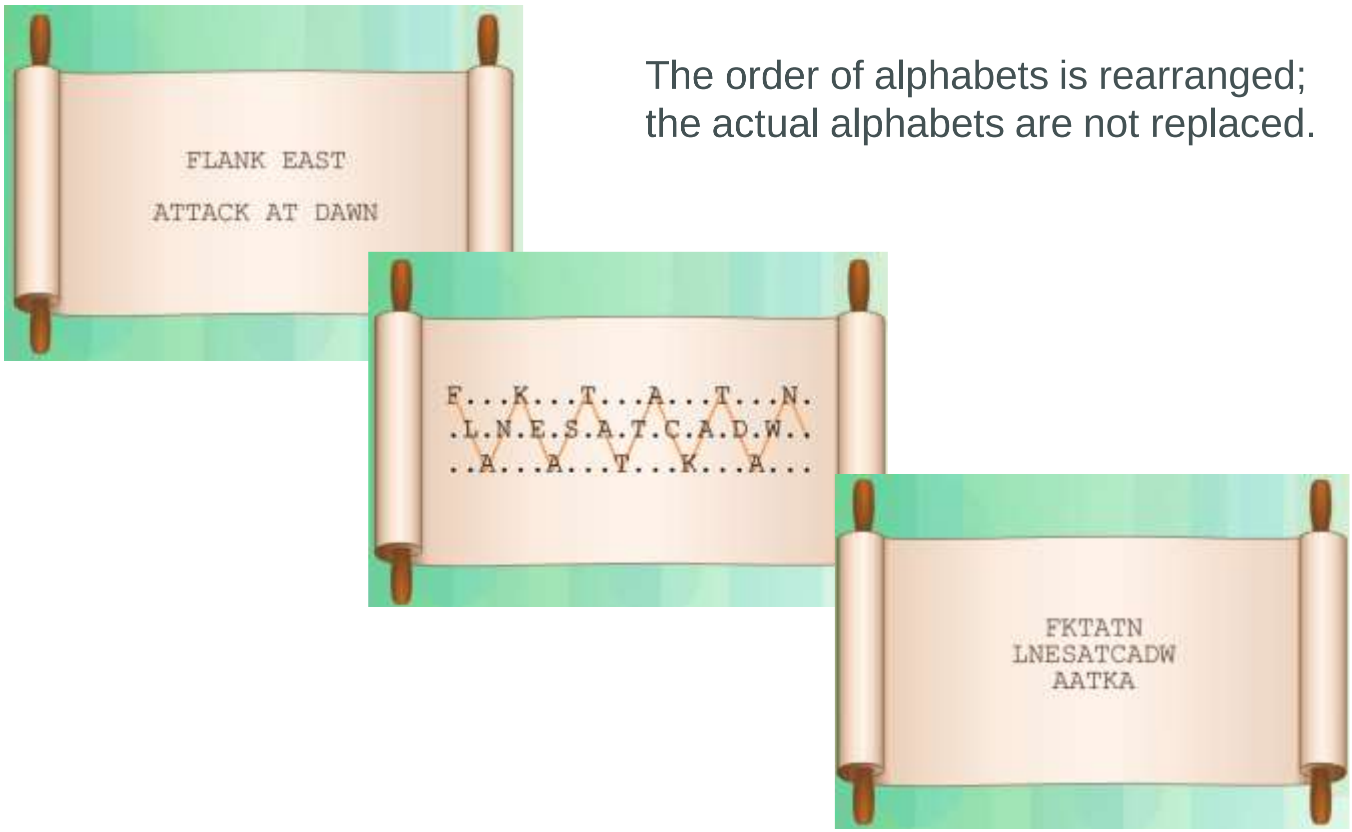
e=000 h=001 i=010 k=011 l=100 r=101 s=110 t=111

Decryption: Ciphertext ⊕ Key = Plaintext

	s	r	l	h	s	s	t	h	s	r
Ciphertext:	110	101	100	001	110	110	111	001	110	101
Key:	111	101	110	101	111	100	000	101	110	000
Plaintext:	001	000	010	100	001	010	111	100	000	101
	h	e	i	l	h	i	t	l	e	r

(Source: Information Security: Principles and Practice)

Transposition Ciphers

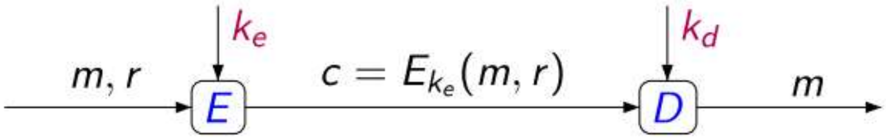


The order of alphabets is rearranged; the actual alphabets are not replaced.

Simple way to see crypto

Three algorithms

- **G** : Key generation
- **E** : Encryption
- **D** : Decryption



- **Ke**: Encryption Key
- **Kd**: Decryption key
- **m**: Message to send
- **r**: receiver

Kerckhoffs Principle

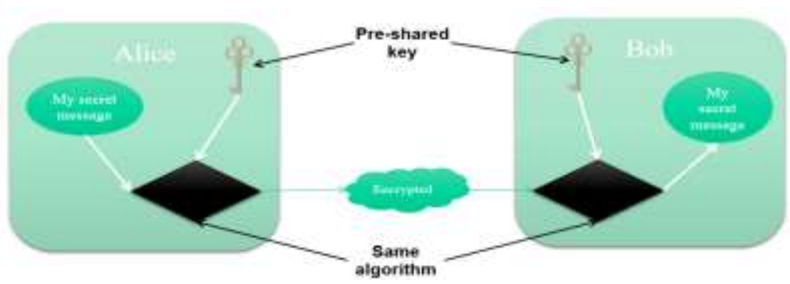
- **Algorithms** should be public but keys should be private

Integrity	Authentication	Confidentiality
MD5	HMAC-MD5	DES
SHA	HMAC-SHA-1	3DES
	RSA and DSA	AES

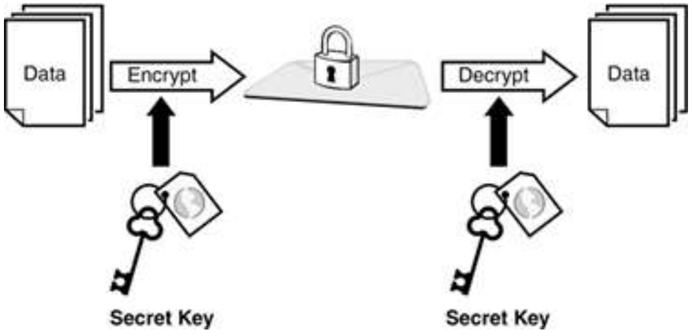
- Opponents argue that both should be private in order to achieve security through obscurity
- **The security relies on the secrecy of the key, not the secrecy of the algorithm**
- In actual fact, the more popular algorithms publicise the algorithm and tend to use the Public Key Infrastructure

Modern Symmetric Cryptography principles

- **Shared key** of **Secret Key**
- Only one Key is used to **encrypt** and **decrypt** the message
- Private key secretly agreed by Alice and Bob
  - **In practice**: **high efficiency** in terms of compu $K_e = K_d = K$
  - **The key must stay a secret**
  - **Challenge**: exchange private keys securely
- Encryption/decryption functions: DES, 3DES, AES, IDEA, RC2/4/5/6, and Blowfish.



Weaknesses of symmetric encryption



- Keys can be compromised (only one needs to be compromised)
- Need mechanisms to **exchange private keys securely**
- Does not implement **non-repudiation**
- **Not scalable** (large groups need equally large number of keys)
- Keys need to be **regenerated** often (when people leave)

Types of symmetric key algorithms

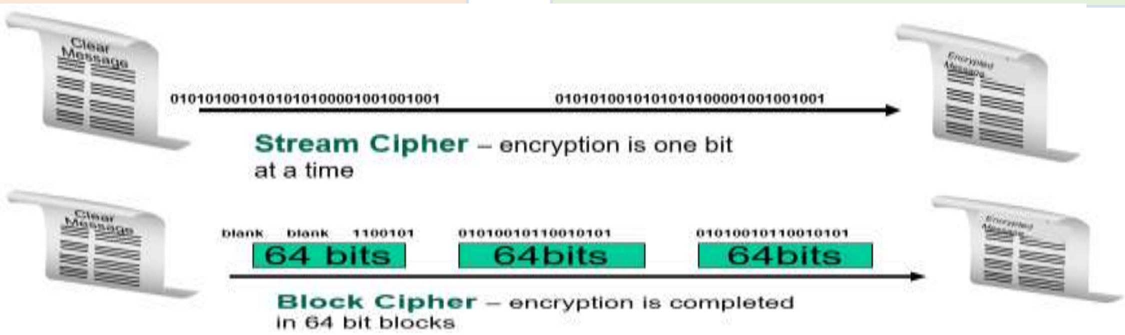
Symmetric-key encryption can use either **stream ciphers** or **block ciphers**:

Stream ciphers

- Encrypt the digits (typically bytes) of a message one at a time.

Block ciphers

- take a number of bits and encrypt them as a single unit, padding the plaintext so that it is a multiple of the block size.
- The Advanced Encryption Standard (AES) algorithm approved by **NIST** uses 128-bit blocks.



Any questions ?